

Extended Runs 2

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New Runs

The next series of runs needs to extend the runs elicited above, plus fill in the energies that are missing. The `InitMDRecord!` function was updated to zero in on the energies in a better way. Previously, the energy was set as a parameter to the random number generation, but the random generation could result in many different energies. The old code was as follows:

```
ss.v .= sqrt( Temp ) .* randn( eltype( ss.v ), ss.N)
```

The new code is as follows:

```
ss.v .= randn( eltype( ss.v ), ss.N)
ke = 0.5 * sum( dot.(ss.v, ss.v) ) / ss.N
ss.v .= sqrt( Temp / ke ) .* ss.v
```

As can be seen, the random numbers are generated first, and then they are scaled to meet the temperature. It is still not exact, as linear and angular momentum are corrected after the kinetic energy generation, and this does change it a bit, but it ought to be a lot closer now.

Runs were made with the following.

```
function runn( td::AbstractString, t::Integer, sname::AbstractString,
              initsteps::Integer=1000, rsteps::Integer=5000, runs::Integer=3)
    s = "0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz"
    temp = t/1000
    @show temp
    tname = @sprintf "%.4d" t
    basename = sname * tname
    @show basename
```

```

cd( td)
io = open( basename * ".txt", "a")
println( io, "Starting Run $basename: $(now())")
startindx = 1
MD2.InitSystem( basename * '.' * s[ startindx], temp=temp, io=io)
println( io, "Starting Run $basename: $(now())")
MD2.RunSystem( basename * '.' * s[ startindx],
               basename * '.' * s[ startindx += 1], initsteps, io=io)
for i in 1:runs
    MD2.RunSystem( basename * '.' * s[ startindx],
                  basename * '.' * s[ startindx += 1], rsteps, io=io)
    println( io, "extension finished: $(now())")
end
close( io)
end

function newruns(td::AbstractString, sname::AbstractString,
                tlist::Vector{<:Integer}, initsteps::Integer, rsteps::Integer,
                runs::Integer = 3)
    Threads.@threads for t in tlist
        runn( td, t, sname, initsteps, rsteps, runs)
    end
end

tlist = [i for i=26:38]
@time newruns( td, "S2b.", tlist, 1000, 5000, 10)

```

This last line creates a 1k initialization run, plus ten 5k runs for a total of 51k steps. The runs were created in the **Data** directory, and will be moved to the 1980 directory when they have been sufficiently processed. I expect very long runs required to achieve equilibrium. The files are processed below.

```

MD2DF.S2b <- MD2DFfile( "MD2DF.S2b.RDS", "H:/Data/DSP/MD/Data",
                       "S2b.*.scalars.csv")

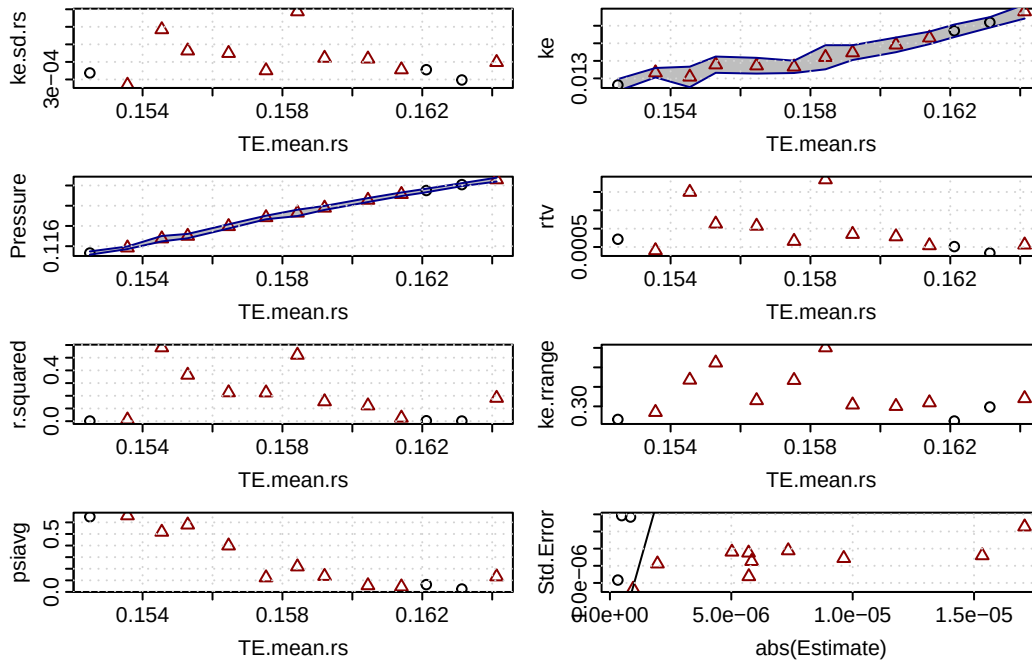
```

That is a new function, created for a variety of reasons. Now, the script can be run, and **createdf** will only be called if there is a new file to be made. Here is a summary of the main parameters:

```

op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,
mgp = c(1.7, 0.4, 0) )
plotScalars( MD2DF.S2b)

```



```
par(op)
```

Only three runs are considered valid, and we ought to be suspicious of these.

```
with( MD2DF.S2b, MD2DF.S2b[Fpvalue > 0.5,
  c("TE.mean.rs", "energy", "steps.avg")])
```

	TE.mean.rs	energy	steps.avg
1	0.1524918	0026	33500
11	0.1621160	0036	33500
12	0.1631320	0037	50000

```
xn26 = combineruns( "H:/Data/DSP/MD/Data", "S2b\\\.0026\\.")
plotMD2( xn26, dt=0.01)
```

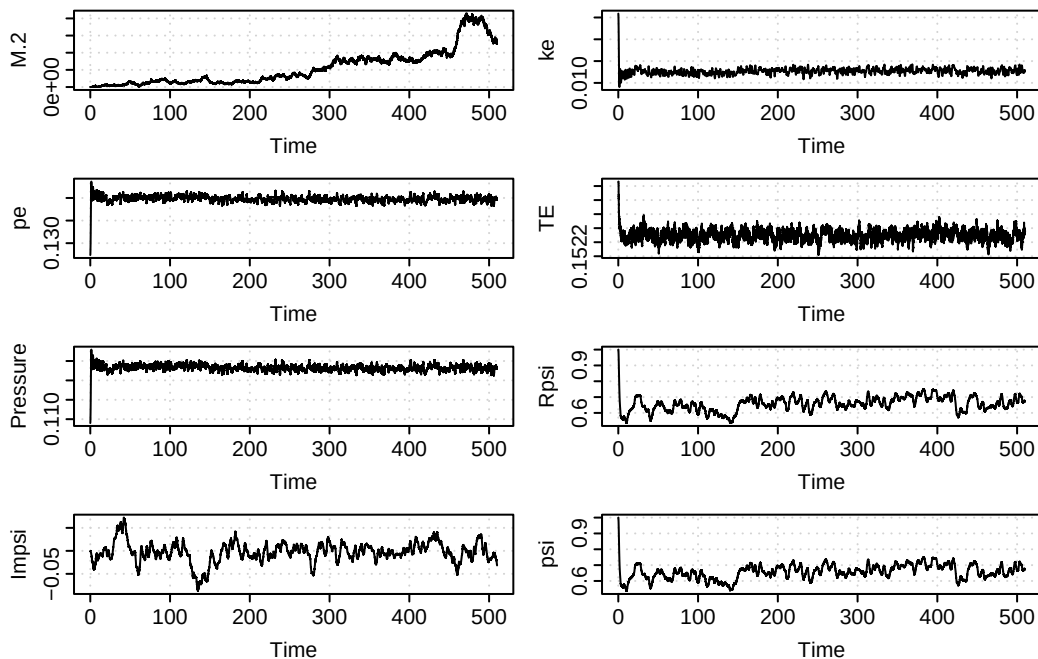


Figure 1: Plots of the basic data in run S2b.0026. Note that this plot was made after further steps were added.

```
xn36 = combineruns( "H:/Data/DSP/MD/Data", "S2b\\.0036\\.")
plotMD2( xn36, dt=0.01)
```

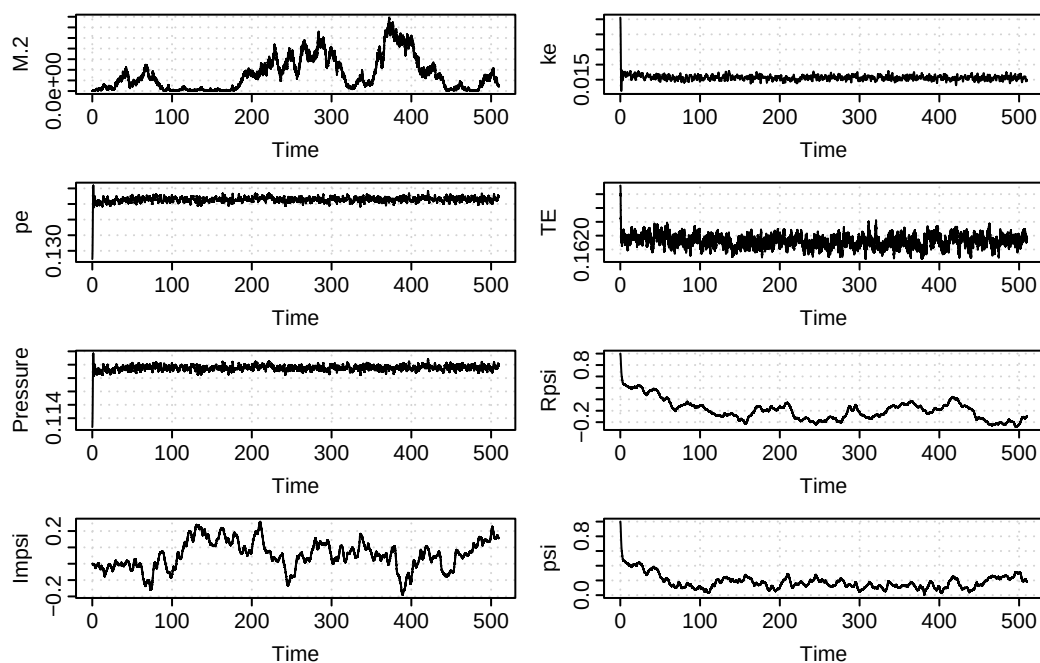



Figure 2: Plots of the basic data in run S2b.0036. Note that this plot was made after further steps were added.

```
xn37 = combineruns( "H:/Data/DSP/MD/Data", "S2b\\.0037\\.")
plotMD2( xn37, dt=0.01)
```

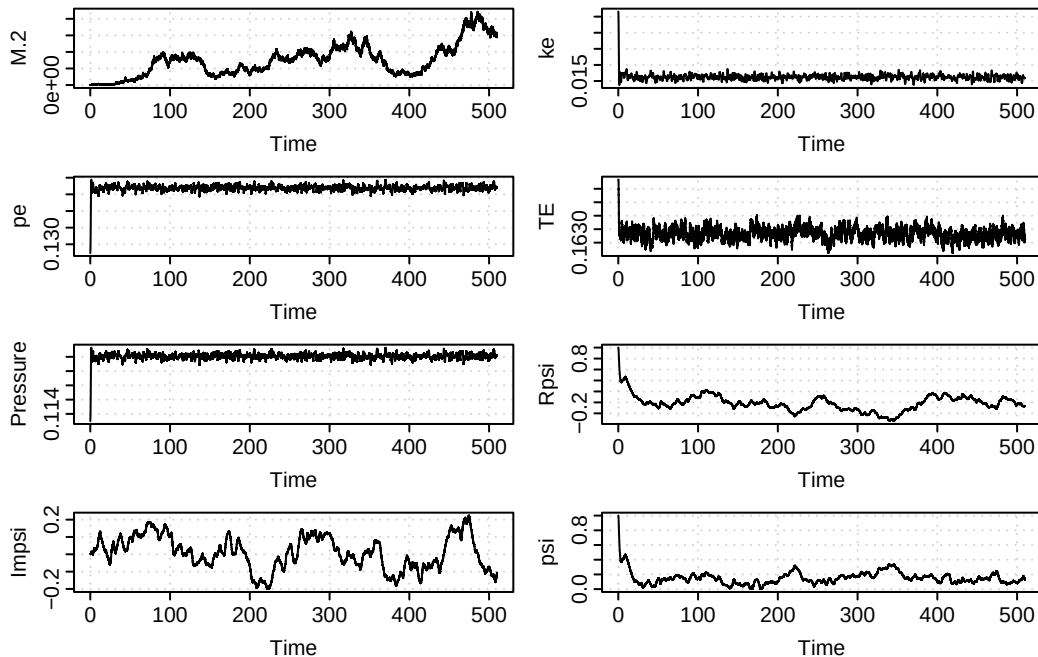


Figure 3: Plots of the basic data in run S2b.0037. Note that this plot was made after further steps were added.

Those three actually look pretty good.

Extending the rest

Ran 20 more runs of the following files: ["0027", "0028", "0029", "0030", "0031", "0032", "0033", "0034", "0035", "0038"] (used `script5.jl` after adding some of the script functions created earlier to the code base). Note that I forgot one file, so it had to be run by itself. Below is the new standard for extending runs.

```
include("C:\\Users\\phils\\Documents\\DSP\\MD\\dev\\MD2\\src\\MD2.jl")
using .MD2

origin = pwd()

td = "H:\\Data\\DSP\\MD\\Data"
cd( td)

list1 = ["0027", "0028", "0029", "0030", "0031", "0032", "0033", "0034", "0035"]
```

```
sname = "S2b."
start = 'B' # Single quotes make it a single char
@time MD2.filerun_thread( td, list1, sname, start, 5000, 20 )

@time MD2.extendrun(td, sname, "0038", 'B', 5000, 20)
```

Note that we are at “U” in the index, with only 31 more to go. The next set of extended runs will have to use larger step blocks.

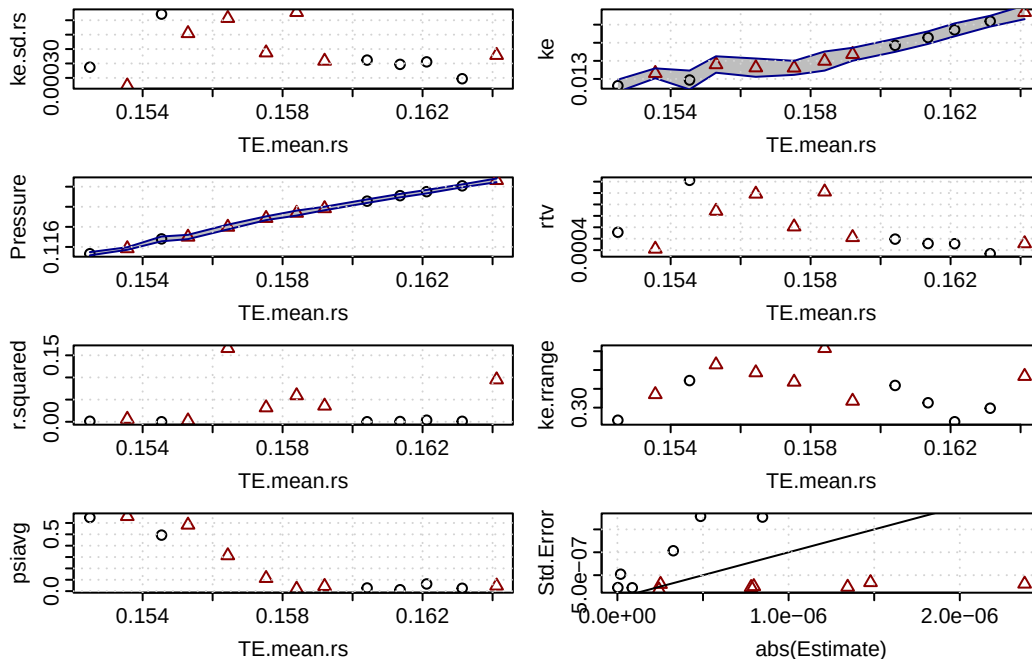
```
MD2DF.S2b.2 <- MD2DFfile( "MD2DF.S2b.2.RDS", "H:/Data/DSP/MD/Data",
                          "S2b.*.scalars.csv")
```

Here are the states of the main scalars. Note that we’ve replaced the TE.mean vs TE.Init with the ke.rrange. This is the relative range of the raw kinetic energy, and is computed by

```
MD2DF$ke.rrange <- (MD2DF$ke.max.raw - MD2DF$ke.min.raw) / MD2DF$ke.mean.rs
```

So we can see the full range of the kinetic energy.

```
op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,
        mgp = c(1.7, 0.4, 0) )
plotScalars( MD2DF.S2b.2)
```



```
par(op)
```

We still have seven runs that are not considered in equilibrium.

```
with( MD2DF.S2b.2, MD2DF.S2b.2[Fpvalue < 0.5,  
  c("TE.mean.rs", "energy", "steps.avg")])
```

	TE.mean.rs	energy	steps.avg
2	0.1535615	0027	150000
4	0.1552966	0029	150000
5	0.1564345	0030	150000
6	0.1575282	0031	150000
7	0.1584004	0032	150000
8	0.1591999	0033	150000
13	0.1641156	0038	150000

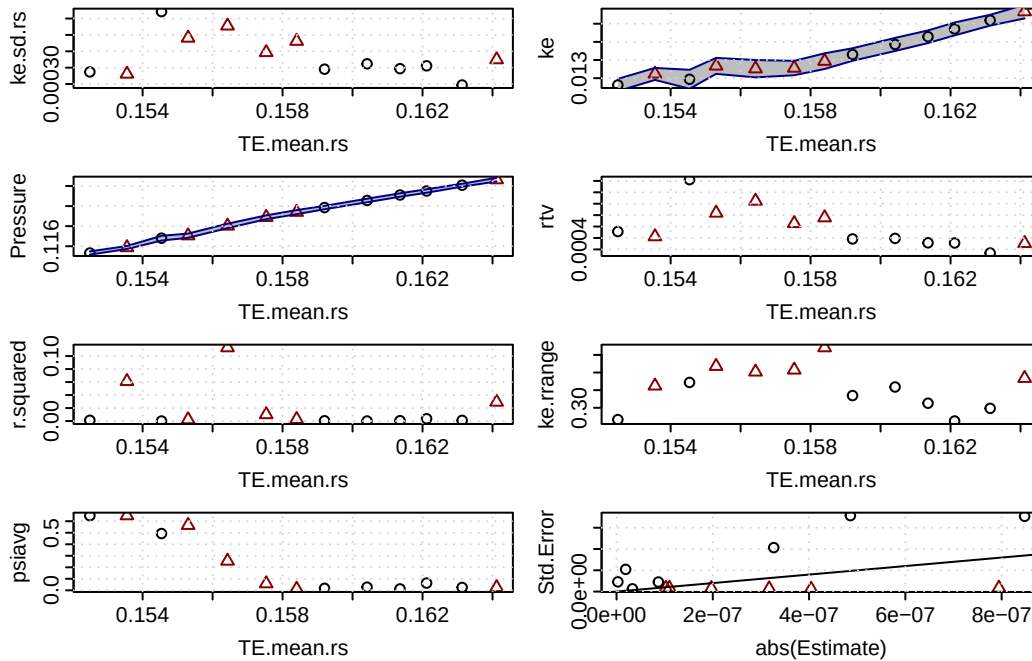
Well, I tested a new feature here. It turns out that lower case letters are not unique to the file system on *Windows*. I think I knew this, but forgot. Fortunately, I had also forgotten that I anticipated such a problem, and made new files with “(n), for example, S2b.00038.a(1).csv as the filename instead of crashing. but now I’m left with figuring out what to do with these files.

I should probably just open them all with `combineruns` and sort the total open series. However, the step starts at 1 in each .csv file. It was foolish to keep the entire state, but not the steps between runs. This requires a change to the MD2 program to keep track of the steps. A 64 bit integer will be enough, as even if the program were a million times faster than the current 0.1s per thousand steps, it would still take 29 years to run out of count space ($0.1 * 2^{63} / (1000000 * 1000 * 3600 * 24 * 365) = 29.5$). Once this is done, future runs can be combined on step number. Until then, I will create a new version of `combineruns` that sorts by date & time using `files <- files[ order( file.mtime( files))]`. After MD2 is updates and a new series of runs are started, `combineruns` will need another update.

```
MD2DF.S2b.3 <- MD2DFfile( "MD2DF.S2b.3.RDS", "H:/Data/DSP/MD/Data",  
  "S2b.*.scalars.csv")
```

And here are the scalars:

```
op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,  
mgp = c(1.7, 0.4, 0) )  
plotScalars( MD2DF.S2b.3)
```



```
par(op)
```

A list of the still “not valid” runs:

```
with( MD2DF.S2b.3, MD2DF.S2b.3[Fpvalue < 0.5,
  c("TE.mean.rs", "energy", "steps.avg")])
```

	TE.mean.rs	energy	steps.avg
2	0.1535524	0027	350000
4	0.1552995	0029	350000
5	0.1564268	0030	350000
6	0.1575323	0031	350000
7	0.1584014	0032	350000
13	0.1641174	0038	350000

Only “0033” reached validity with the additional 20k points. One thing odd: point “0038” remains “not valid” while the five cooler points 33 through 37 are considered valid. It bears looking into.

```
xn38 = combineruns( "H:/Data/DSP/MD/Data", "S2b\\.0038\\.")
plotMD2( tail( xn38, -1000), dt=0.01)
```

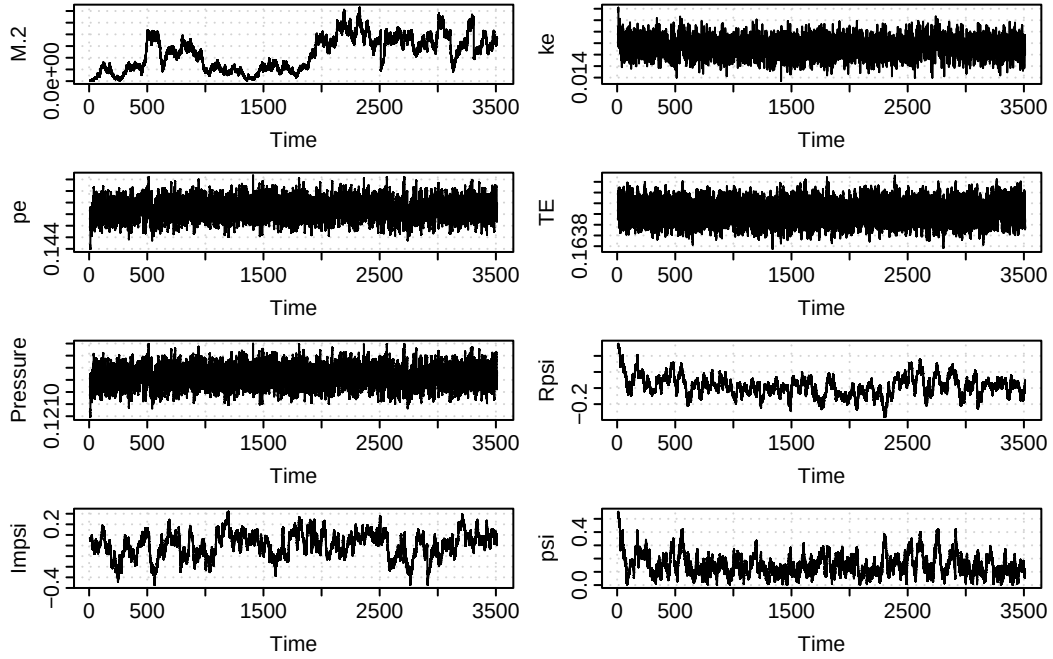
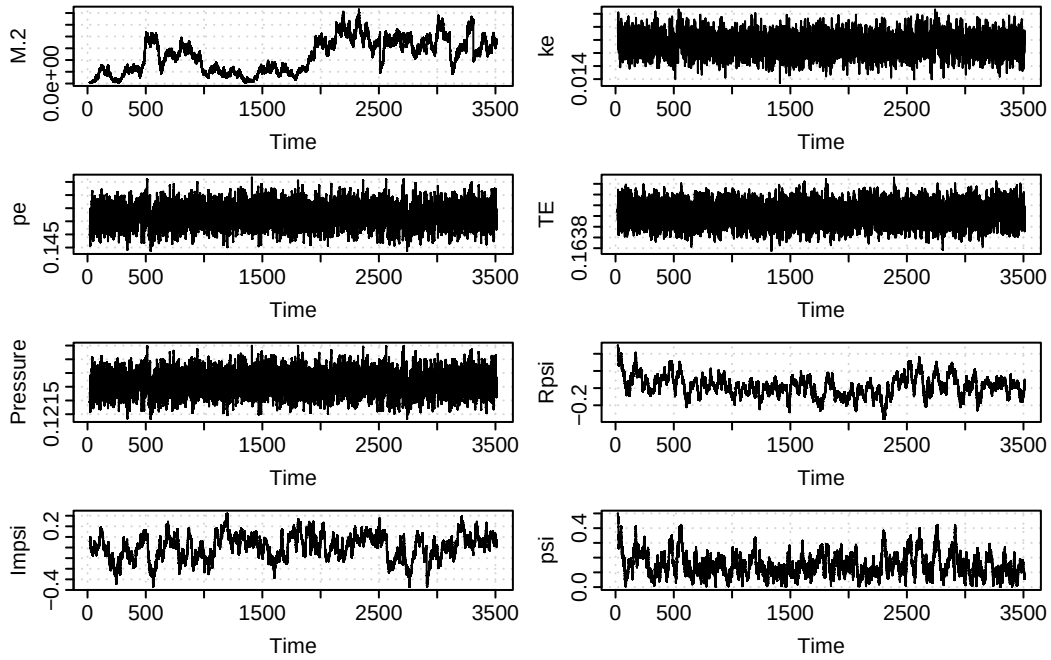


Figure 4: Plots of the basic data in run S2b.0038.

```
plotMD2( tail( xn38, - 2000), dt=0.01)
```



```

navg <-1000
skip <- 10000
(t <- procMD2DF( xn38, skip=skip, rsblk=navg))

```

sdf: Total Size: 341000, Step: 2048, Number of FFTs: 165, window Power: 4096

total.steps	steps.avg	TE.init	xMomentum.mean.rs
3.510000e+05	2.280000e+05	1.649442e-01	-1.225511e-10
xMomentum.sd.rs	yMomentum.mean.rs	yMomentum.sd.rs	M.2.mean.rs
9.807743e-11	-1.074947e-10	5.027784e-11	3.878810e-20
M.2.sd.rs	ke.mean.rs	ke.sd.rs	pe.mean.rs
2.714311e-20	1.665457e-02	2.756615e-04	1.474610e-01
pe.sd.rs	TE.mean.rs	TE.sd.rs	Pressure.mean.rs
2.778123e-04	1.641155e-01	3.877352e-05	1.226470e-01
Pressure.sd.rs	Rpsi.mean.rs	Rpsi.sd.rs	Impsi.mean.rs
1.347464e-04	1.806550e-03	1.094161e-01	4.610792e-03
Impsi.sd.rs	psiabs.mean.rs	psiabs.sd.rs	xMomentum.min.raw
1.116958e-01	1.454436e-01	6.713325e-02	-3.123899e-10
xMomentum.max.raw	yMomentum.min.raw	yMomentum.max.raw	M.2.min.raw
1.067583e-10	-2.305718e-10	4.285883e-11	1.082856e-21
M.2.max.raw	ke.min.raw	ke.max.raw	pe.min.raw
1.258258e-19	1.370005e-02	1.932256e-02	1.447334e-01
pe.max.raw	TE.min.raw	TE.max.raw	Pressure.min.raw
1.503309e-01	1.637735e-01	1.644450e-01	1.213251e-01
Pressure.max.raw	Rpsi.min.raw	Rpsi.max.raw	Impsi.min.raw
1.239975e-01	-3.602424e-01	4.129934e-01	-3.725171e-01
Impsi.max.raw	psiabs.min.raw	psiabs.max.raw	nr
3.199893e-01	1.186801e-03	4.228041e-01	2.280000e+02
Estimate	Std.Error	r.squared	Fval
1.747615e-07	2.798466e-07	1.722639e-03	3.899883e-01
Fpvalue	df.dendf	rtv	psiavg
5.329355e-01	2.260000e+02	2.739589e-04	4.952073e-03
specwidth			
2.929688e-02			

```

rxn38ke <- rsmavg( tail( xn38$ke, -skip), navg)
rxn38 <- data.frame( ke=rxn38ke,
                    step=seq( from=navg/2, by=navg, along.with=rxn38ke))
lmrxn38 <- lm(ke ~ step, data = rxn38) # fit across entire data set
(slmrxn38 <- summary( lmrxn38))

```

Call:

```
lm(formula = ke ~ step, data = rxn38)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.917e-04	-1.870e-04	5.500e-07	1.646e-04	1.043e-03

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.669e-02	3.122e-05	534.696	<2e-16 ***
step	-2.027e-10	1.586e-10	-1.279	0.202

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0002882 on 339 degrees of freedom

Multiple R-squared: 0.004799, Adjusted R-squared: 0.001863

F-statistic: 1.635 on 1 and 339 DF, p-value: 0.2019

```
#lenrx31 <- nrow( rx31)
#third <- floor(lenrx31 / 3) # we'll exclude the least number of points
trxn38 <- tail( rxn38, t['nr']) # procMD2DF fit
sstrxn38 <- stat.summary( trxn38$ke)
lmtrxn38 <- lm( ke ~ step, data=trxn38) # fit to tail
summary( lmtrxn38)
```

Call:

```
lm(formula = ke ~ step, data = trxn38)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.574e-04	-1.724e-04	-8.180e-06	1.550e-04	1.031e-03

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.660e-02	6.614e-05	251.033	<2e-16 ***
step	1.748e-10	2.798e-10	0.624	0.533

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0002781 on 226 degrees of freedom

Multiple R-squared: 0.001723, Adjusted R-squared: -0.002695
F-statistic: 0.39 on 1 and 226 DF, p-value: 0.5329

```
#ftlmxn38 <- lm( ke ~ step, data=head( rxn38, -t['nr'])) # Fit to head of data.
plot( tail( xn38$ke, -skip), type='l', panel.first=grid()) # plot raw data
points(ke ~ step, data=rxn38, col='darkred') # Plot all resample points
abline( coef=coef( lmrxn38)) # Fit to all resample points
points(ke ~ step, data=trxn38, col='blue') # Plot tail resample points
abline( coef=coef( lmtrxn38), col='blue') # Fit to the tail resample points
abline( h=sstrxn38["mean"], col='darkgreen')
abline( h=sstrxn38["min"], col="darkgreen", lty=2)
abline( h=sstrxn38["max"], col="darkgreen", lty=2)
#abline( coef=coef( ftlmxn38), col='darkred')
```

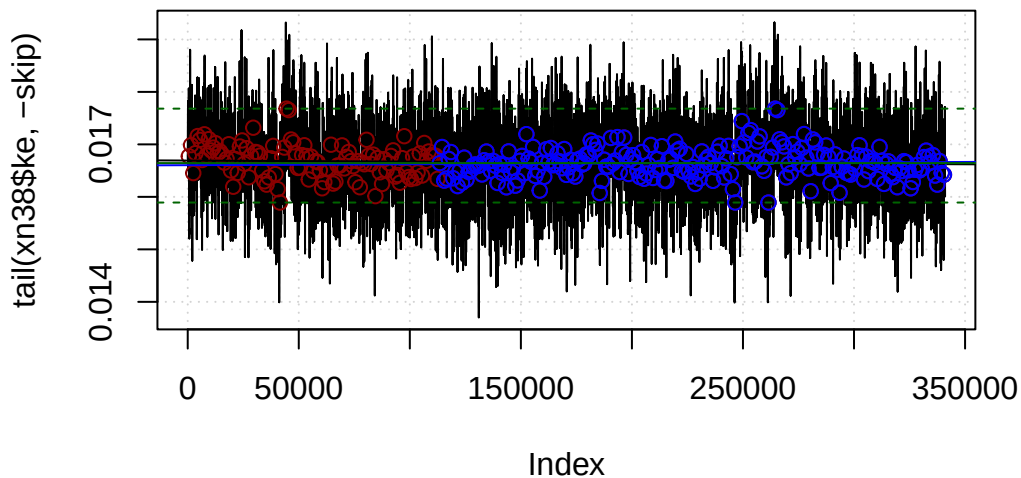


Figure 5: S2.0031 (Figure 4) kinetic energy (less the first 2,000 points) with fits to the resampled data, the black line is the fit to all of the data (red and blue circles), and the last two-thirds of truncated run S2-0031 (blue circles and line). Green lines indicate the mean (solid), and min & max (dashed) of the blue points. The red line is the regression line for the red points.

Made substantial changes to filename propagation.

```

td = "H:\\Data\\DSP\\MD\\Data"
list1 = "S2B." .* ["0027", "0028", "0029", "0030", "0031", "0032", "0034", "0035"]
"S2B.0027"
"S2B.0028"
"S2B.0029"
"S2B.0030"
"S2B.0031"
"S2B.0032"
"S2B.0034"
"S2B.0035"
MD2.filerun_thread(td, list1, ".e(1)", 20000, 35)

```

This did result in one error: nested task error: SystemError: opening file "S2B.0028.e(1).jld2": No such file or directory. Also, flouting the case insensitivity failed with R, and I had to rename the "S2B*.csv" files using Microsoft PowerTool's `PowerRename`. There were also errors in the list: the 0028 run should not have been in the list (thus the error), and the 0038 run *should* have been in the list, and was not.

```

MD2DF.S2b.4 <- MD2DFfile( "MD2DF.S2b.4.RDS", "H:/Data/DSP/MD/Data",
                          "S2*.scalars.csv", verbose=TRUE)

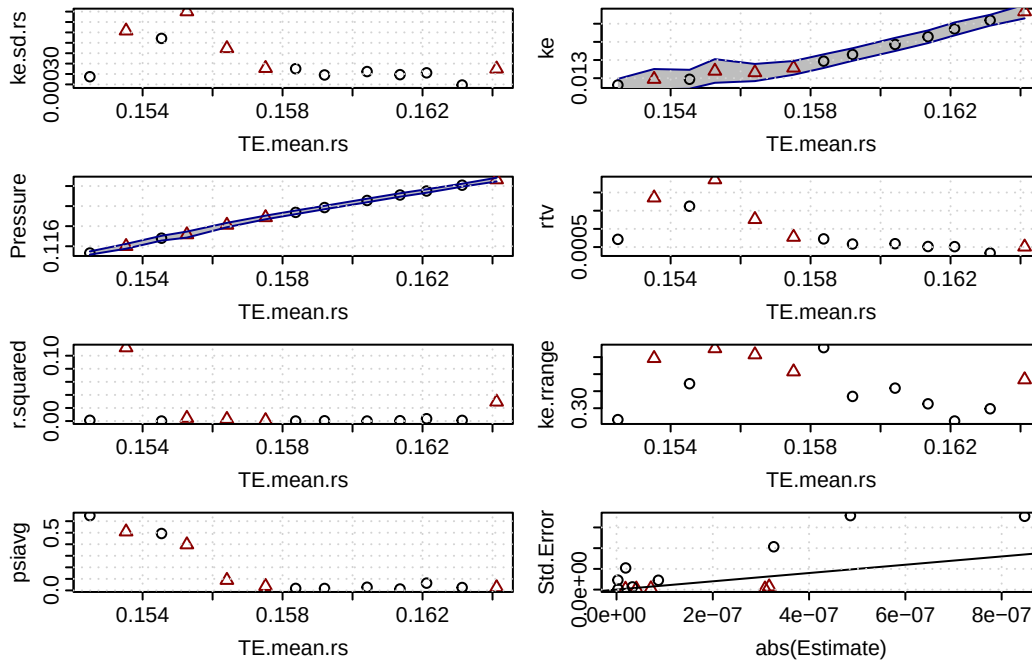
```

And here are the scalars:

```

op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,
mgp = c(1.7, 0.4, 0) )
plotScalars( MD2DF.S2b.4)

```



```
par(op)
```

A list of the still “not valid” runs:

```
with( MD2DF.S2b.4, MD2DF.S2b.4[Fpvalue < 0.5,
  c("TE.mean.rs", "energy", "steps.avg")])
```

	TE.mean.rs	energy	steps.avg
2	0.1535270	0027	1050000
4	0.1552654	0029	1050000
5	0.1564087	0030	1050000
6	0.1575144	0031	1050000
13	0.1641174	0038	350000

So only the 0032 run became valid. There are two oddities here: runs 0028 and 0038. 0028 because it became valid after “only” 151k steps (the last 100k being used) while the surrounding energies remain un-valid at over 1 million steps. 0038 because it remains on the not-valid list at 351k steps, while two lower energies were validated at only 51k steps.

Melting!

In the process of looking at S2b.0027, an error in `combineruns` was discovered and fixed. This tends to invalidate the prior equilibrium regressions, so the MD2DFdata must be re-compiled. Nevertheless, the following appears correct.

```
xn27 = combineruns( "H:/Data/DSP/MD/Data", "S2b\\.0027\\. ", verbose=TRUE)
```

Searching Directory: H:/Data/DSP/MD/Data for pattern: S2b\\.0027\\..(.*).scalars.csv

Files found: H:/Data/DSP/MD/Data/S2b.0027.0.scalars.csv H:/Data/DSP/MD/Data/S2b.0027.1.scalars.csv

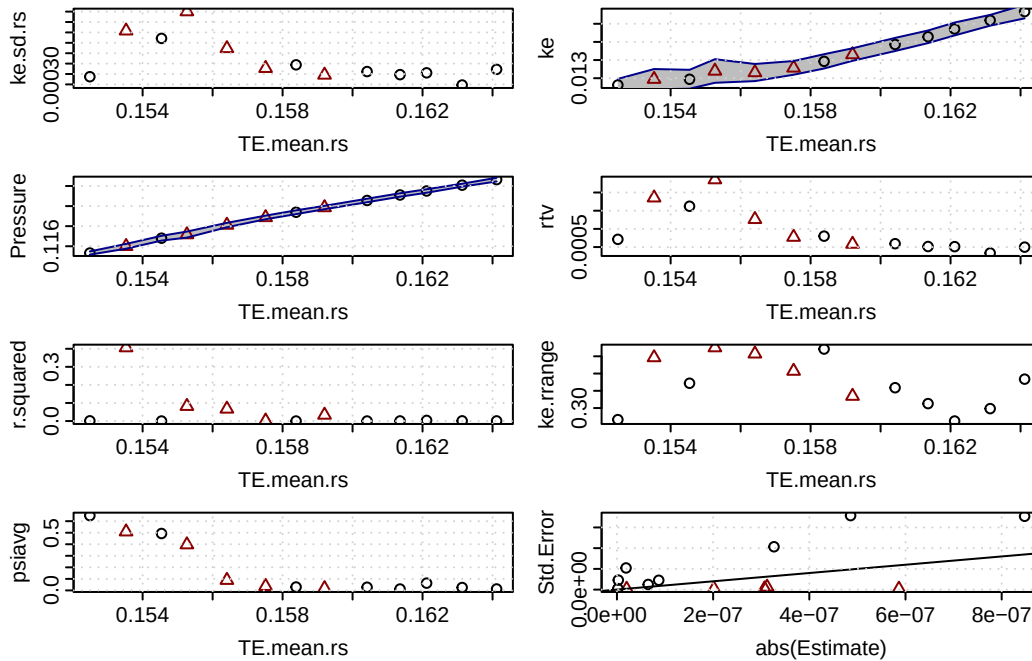
Just after $6,000\tau$, ψ starts to move. At the same time, the potential energy starts to increase and the kinetic energy starts to decrease. This is much like putting salt on ice: the temperature decreases as it melts.

All the Data

Rerunning

```
MD2DF.S2b.5 <- MD2DFfile( "MD2DF.S2b.5.RDS", "H:/Data/DSP/MD/Data",  
                          "S2*.scalars.csv", verbose=TRUE)
```

```
op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,  
mgp = c(1.7, 0.4, 0) )  
plotScalars( MD2DF.S2b.5)
```

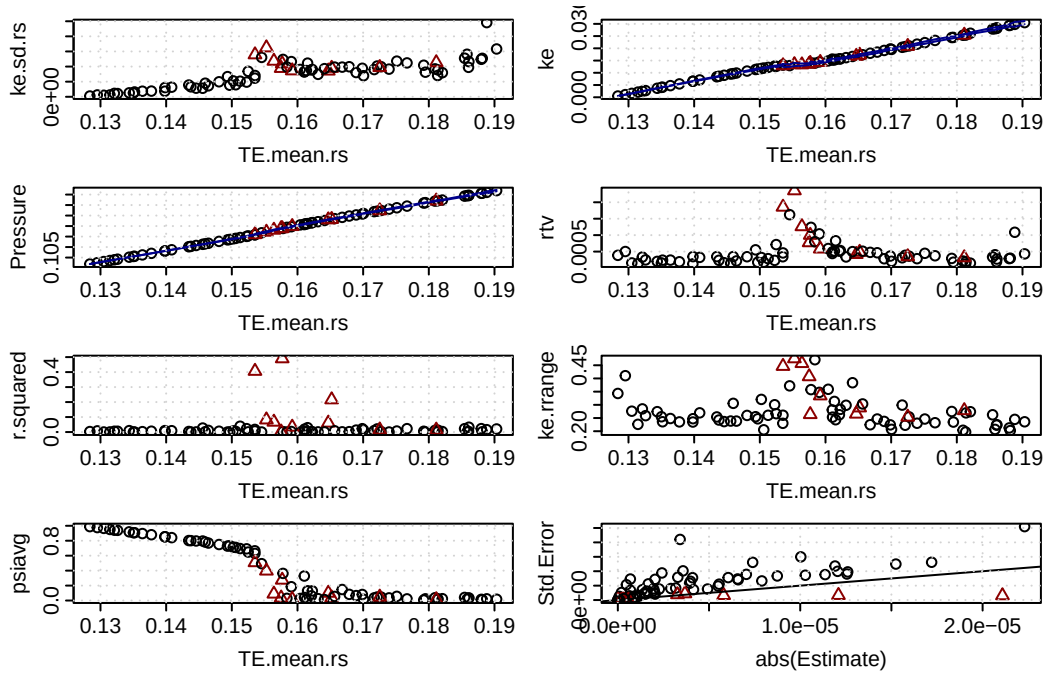


```
par(op)
```

```
load("MD2DF.4.S2.Rdata")
MD2DF.t <- rbind( MD2DF, MD2DF.S2b.5)
```

There are a lot of hot runs in there, so we'll filter for total energies below 0.19.

```
op= par( mfrow = c(4, 2), mar = c(3, 3, 0.5, 0.5), tcl = -0.3,
         mgp = c(1.7, 0.4, 0))
with( MD2DF.t, plotScalars( MD2DF.t[TE.mean.rs < 0.191,]))
```



```
par(op)
```

Here are the runs that still need to be extended. Note that 0028 should remain suspect, as it has a slightly higher energy than 0027 that literally melted before our eyes.

```
with( MD2DF.t, MD2DF.t[Fpvalue < 0.5,
  c("series", "energy", "TE.mean.rs", "steps.avg")])
```

	series	energy	TE.mean.rs	steps.avg
30	S2	0030	0.1576751	25000
37	S2	0037	0.1652028	25000
38	S2	0038	0.1646941	25000
51	S2	0051	0.1725277	25000
59	S2	0059	0.1811328	25000
74	S2	0074	0.2060394	25000
102	S2b	0027	0.1535270	1050000
104	S2b	0029	0.1552654	1050000
105	S2b	0030	0.1564087	1050000
106	S2b	0031	0.1575144	1050000
108	S2b	0033	0.1591992	350000